🔷 16.11.2 Core Dynamics Consistency (Final — Integrated Version)

$$\Psi_{k+1} = \Psi_k + dt,F(\Psi_k, L_k, P_k, u_k) + \sigma\sqrt{dt},\xi_k$$

with:

$$F(\Psi_k) =
-\nabla \mathcal{E}(\Psi_k)
- L_k \Psi_k
+ \alpha(PAC_k)\nabla \ln P_k
+ B(u_k)$$

Consistency Properties

Under the assumptions established in Sections 16.1–16.8:

- Boundedness

$$\|F(\Psi_k)\| \le C$$

- Global Lipschitz Continuity

$$\|F(x) - F(y)\| \le L_{\mathrm{tot}} \|x - y\|$$

- Well-Posedness

$$\exists!\ \Psi_k$$

- Noise Boundedness

$$\mathbb{E}\|\sigma\sqrt{dt}\,\xi_k\|^2 = \sigma^2 dt$$

- Closed-Loop Structure

$u_k = \pi(\Psi_k)$

Implication

Thus:

the system dynamics are bounded, Lipschitz continuous, well-posed, and fully closed-loop consistent, forming a valid foundation for the global evolution law in Section 16.11.

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